

Divisibility, Euclidean Algorithm, Greatest Common Divisor, Bezout, and Congruences

N-20221031

§1.1 Divisibility

The first page begins with divisibility. For integers a, b , with $a \neq 0$, we write

$$a \mid b$$

if there exists $k \in \mathbb{Z}$ such that

$$b = ak.$$

If no such integer exists, we write $a \nmid b$.

The basic properties recorded in the note are:

- (i) $a \mid a$ and $a \mid 0$;
- (ii) if $a \mid b$ and $b \mid a$, then $a = \pm b$;
- (iii) if $a \mid b$ and $b \mid c$, then $a \mid c$;
- (iv) if $a \mid b$ and $a \mid c$, then $a \mid bx + cy$ for all integers x, y ;
- (v) if $a \mid b$, then $ac \mid bc$ for every integer c .

The handwritten proofs repeatedly use the definition. For instance, if $b = am$ and $c = an$, then

$$bx + cy = a(mx + ny),$$

so $a \mid bx + cy$.

§1.2 Division algorithm

The division algorithm says: if $a, b \in \mathbb{Z}$ and $b > 0$, then there exist unique integers q, r such that

$$a = bq + r, \quad 0 \leq r < b.$$

The note calls attention to uniqueness. If

$$a = bq + r = bq' + r', \quad 0 \leq r, r' < b,$$

then

$$b(q - q') = r' - r.$$

The right side has absolute value less than b , while the left side is a multiple of b . Therefore $r = r'$ and then $q = q'$.

§1.3 Greatest common divisor and least common multiple

The greatest common divisor is written

$$(a, b),$$

and the least common multiple is written

$$[a, b].$$

The note records the identity

$$(a, b)[a, b] = |ab|.$$

It also records useful gcd properties:

$$(a, 1) = 1, \quad (a, 0) = |a|, \quad (a, a) = |a|,$$

$$(a, b) = (b, a) = (ta, tb)/|t|,$$

and most importantly the Euclidean step

$$(a, b) = (a - b, b),$$

or more generally

$$(a, b) = (a - qb, b).$$

The proof of the Euclidean step is inclusion both ways. Any common divisor of a and b divides $a - qb$. Conversely, any common divisor of $a - qb$ and b divides a .

§1.4 Euclidean algorithm

Repeatedly applying

$$(a, b) = (b, r), \quad a = bq + r,$$

produces the Euclidean algorithm. The last nonzero remainder is the gcd.

The note's red proof makes the descent explicit: the remainders strictly decrease and remain nonnegative, so the process must terminate. This is not just an algorithm; it is a proof that the gcd can be computed by repeated division.

§1.5 Bezout theorem

Theorem 1.5.1 (Bezout identity)

For integers a, b , there exist integers x, y such that

$$ax + by = \gcd(a, b).$$

Bezout's theorem says that if $a, b \in \mathbb{Z}$ are not both zero, then there exist $s, t \in \mathbb{Z}$ such that

$$as + bt = (a, b).$$

The note proves this by induction on $a + b$ and also implicitly through the Euclidean algorithm. When the gcd is obtained as the last nonzero remainder, one substitutes backward to express that remainder as an integer linear combination of a and b .

A key special case is:

$$(a, b) = 1 \iff \exists s, t \in \mathbb{Z}, as + bt = 1.$$

This is the form most often used in proofs.

§1.6 Gauss lemma and divisibility consequences

Lemma 1.6.1 (Gauss lemma)

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

The next page proves Euclid's lemma or Gauss's lemma:

$$(a, b) = 1, \quad a \mid bc \implies a \mid c.$$

Using Bezout, choose s, t such that

$$as + bt = 1.$$

Multiplying by c gives

$$asc + btc = c.$$

Since $a \mid asc$ and $a \mid bc$ implies $a \mid btc$, we get $a \mid c$.

This is the standard tool behind unique factorization and many modular arguments.

§1.7 Prime numbers

A prime number p satisfies: if

$$p \mid ab,$$

then

$$p \mid a \quad \text{or} \quad p \mid b.$$

This follows from Gauss's lemma. If $p \nmid a$, then $(p, a) = 1$, so from $p \mid ab$ we get $p \mid b$.

The note also uses the property repeatedly in exercises involving prime powers and factorization.

§1.8 Congruences

For an integer $m > 0$, write

$$a \equiv b \pmod{m}$$

if $m \mid (a - b)$. Congruence is compatible with addition and multiplication:

$$a \equiv b, \quad c \equiv d \pmod{m} \implies a + c \equiv b + d \pmod{m},$$

$$ac \equiv bd \pmod{m}.$$

The handwritten examples include reductions of powers modulo m and using congruences to rule out equations.

§1.9 Exercises and common methods

The printed exercise pages ask one to prove identities such as

$$(a, b) = (a - b, b), \quad (a^m - 1, a^n - 1) = a^{(m,n)} - 1,$$

and divisibility statements involving expressions like

$$2^{2n+1} + 1, \quad 3^{2n} - 1.$$

The repeated methods are:

- (i) apply the Euclidean algorithm to gcd expressions;
- (ii) use Bezout when coprimality appears;
- (iii) factor expressions like $x^m - y^m$;
- (iv) reduce powers modulo a suitable modulus;
- (v) use prime factorization when divisor functions appear.

§1.10 From example to method

The note develops number theory from one simple relation: divisibility. The division algorithm gives remainders; remainders give the Euclidean algorithm; the Euclidean algorithm gives Bezout; Bezout gives Gauss's lemma; Gauss's lemma supports prime factorization and congruence arguments. This chain is the backbone of elementary number theory.

§1.11 Division algorithm as normal form

For integers a, b with $b > 0$, division gives

$$a = qb + r, \quad 0 \leq r < b.$$

The remainder r is a normal form for the congruence class of a modulo b .

§1.12 Euclidean algorithm and invariant gcd

The key invariant is

$$\gcd(a, b) = \gcd(b, a - qb).$$

Each Euclidean step preserves the gcd while reducing size. Back-substitution then expresses the gcd as a Bezout combination.

§1.13 Gauss lemma and prime ideals

Lemma 1.13.1 (Gauss lemma, ideal form)

For a prime p , the ideal $(p) \subset \mathbb{Z}$ is prime; equivalently, $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Gauss's lemma says if p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$. In ideal language, (p) is a prime ideal of $\mathbb{Z}$. This is the algebraic property that makes unique factorization work.

§1.14 Normal forms, gcd invariants, and prime divisibility

Divisibility, gcd, Bezout, congruence, and primes form one system. Remainders are quotient normal forms, the Euclidean algorithm preserves gcd, and prime divisibility is the ideal-theoretic foundation of unique factorization.

§1.15 $\mathbb{Z}$ as a PID

The full chain of gcd, lcm, Bezout identity, congruence, and modular inverse is unified by the fact that $\mathbb{Z}$ is a principal ideal domain. Every ideal is of the form

$$(n) = n\mathbb{Z}.$$

For two integers,

$$(a) + (b) = (\gcd(a, b)), \quad (a) \cap (b) = (\text{lcm}(a, b)).$$

This gives a structural proof of many gcd-lcm identities.

For example,

$$\gcd(a, b) \text{lcm}(a, b) = |ab|$$

is the numerical reflection of how prime valuations take minima and maxima.

§1.16 Modular inverses as units

The congruence equation

$$ax \equiv 1 \pmod{n}$$

asks whether the residue class of a is a unit in the ring $\mathbb{Z}/n\mathbb{Z}$. It has a solution iff $\gcd(a, n) = 1$.

More generally,

$$ax \equiv b \pmod{n}$$

has a solution iff $\gcd(a, n) \mid b$. This is because the image of multiplication by a on $\mathbb{Z}/n\mathbb{Z}$ has size $n/\gcd(a, n)$ and consists exactly of multiples of that gcd class.

§1.17 Smith normal form over integers

Linear equations over $\mathbb{Z}$ require more than rank. For an integer matrix A , Smith normal form says there exist unimodular integer matrices P, Q such that

$$PAQ = \text{diag}(d_1, \dots, d_r, 0, \dots, 0), \quad d_i \mid d_{i+1}.$$

This diagonal form solves integer linear systems and classifies finitely generated abelian groups.

Thus gcd and Bezout operations are the 1×1 and row-column-operation shadows of Smith normal form.

§1.18 CRT as product decomposition

When m, n are coprime,

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

The Chinese remainder theorem is therefore not simply a trick for solving congruences. It says the quotient ring modulo a product of coprime moduli decomposes as a product of quotient rings.

This decomposition explains why arithmetic functions and congruence problems split prime by prime.

§1.19 Solving congruences by image and kernel

Multiplication by a defines a group homomorphism

$$m_a : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}, \quad x \mapsto ax.$$

The congruence $ax \equiv b \pmod{n}$ asks whether b lies in the image of this map. The kernel has size $\gcd(a, n)$, so the image has size $n/\gcd(a, n)$.

The solvability condition $\gcd(a, n) \mid b$ is therefore rank-nullity for a finite abelian group. Once solvable, there are exactly $\gcd(a, n)$ solutions modulo n .

§1.20 Smith normal form and congruences

A system of integer linear equations

$$Ax = b$$

can be simplified by invertible integer row and column operations. Smith normal form diagonalizes A into invariant factors d_i . The system reduces to independent divisibility conditions

$$d_i y_i = c_i.$$

Thus the one-variable congruence theorem is the first case of a full normal-form theory for homomorphisms of finitely generated abelian groups.